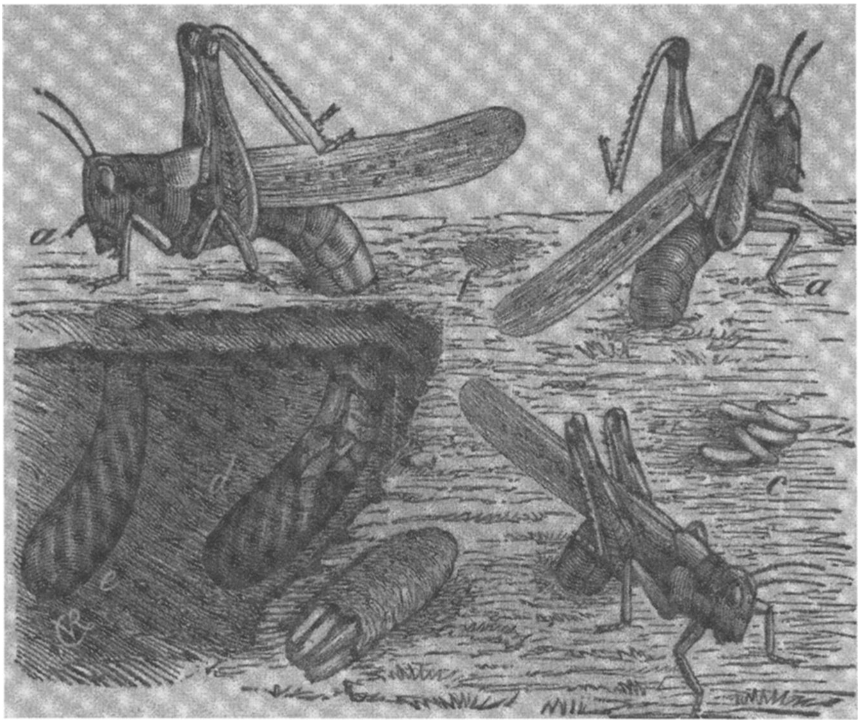

Female locusts depositing eggs. Illustration from "The Rocky Mountain locust, or grasshopper of the West," in *Report of the Commissioner of Agriculture for the year 1877* (Washington, D.C.: Government Printing Office, 1877), plate 1, figure 1 (after Riley).

The Rocky Mountain Locust in Texas

By Stanley D. Casto*

Listen all who live in the land. Has anything like this ever happened in your day or in the days of your forefathers? *(Joel 1:2)*

The first Texans to experience the ravages of the Rocky Mountain locust (*Melanoplus spretus*) must certainly have recalled the question asked by the prophet Joel during the locust plague in ancient Israel. Has anything like this ever happened in your day or in the days of your forefathers? From where did this pestiferous insect come? Could its sudden appearance in such enormous numbers be rationally explained or was it sent, as suggested by Joel, as a reminder that the day of the Lord was near? And of immediate concern, how might the gardens, orchards, and crops be protected from this insatiable destroyer of plant life? These questions would eventually be answered but not before the citizens of Texas had endured a number of devastating visitations.

The Rocky Mountain locust was a migratory grasshopper, now presumed extinct, that lived in the valleys of the Rocky Mountains of the western United States and southern Canada. Large swarms periodically left the place of their nativity and migrated onto the Great Plains of the United States and southern Canada. Although swarming and migration undoubtedly occurred for hundreds of years, it was not until the early 1800s that these phenomena were first noted in the historical literature. The magnitude of the swarms and the damage they caused peaked during the 1870s. The migrations then diminished over the next two decades, and by the turn of the century had

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ceased. So catastrophic was the damage during the 1870s that the federal government created a special entomological commission to study the pest and devise methods for its control. The damage caused by the Rocky Mountain locust in Texas was less severe than that which occurred in regions farther north.[1] The sudden and unpredictable occurrence of the locusts was, however, a source of apprehension and the subject of speculation in the news media of the time. Scientific expertise and governmental assistance were nonexistent, and farmers were forced to devise their own strategies for dealing with the pest. This paper chronicles the invasions of the Rocky Mountain locust, its effect on agriculture, and the struggles of Texans compelled to deal with this once serious but now apparently extinct insect pest.

The swarms of locusts that invaded Texas usually arrived from the north during October or early November, and over the next several weeks deposited enormous numbers of eggs in the soil before dying. Eggs laid in the fall hatched from January through March of the following year. The wingless juveniles were at first sedentary but later congregated into massive schools or armies that migrated over the countryside destroying all vegetation in their path. When mature and able to fly, the locusts gathered into large swarms and departed, usually flying in a northwesterly direction as they passed overhead.

The first visitation of the Rocky Mountain locust to Texas is believed to have occurred in 1845. This conclusion is based on the sources consulted during the early 1870s by C. V. Riley while he was serving as the state entomologist of Missouri.[2] Riley's informants were presumably eyewitnesses to the events of 1845, but no details are given on when or where the locusts appeared or the extent of the damage caused. If locusts did occur in Texas during 1845, it is likely that the swarms descended on the outlying prairies far from major settlements.

The earliest contemporary record of locusts that has been found is in an old Caldwell County deed book, which contains an entry noting that on October 1, 1847, the "Grasshoppers came, remained all fall [but did] not [do] much damage."[3] Nothing was known at that time of the natural history

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[1] Locusts are specialized grasshoppers that have two phases in their life cycle—a solitary phase in which they are sedentary and a gregarious phase in which they are migratory. Locusts were reported in Manitoba, Canada, during 1800 and 1808. They were again seen during 1818 and 1819 in Minnesota and Manitoba. Swarms appeared during 1820 in western Missouri and perhaps in Kansas and regions farther north. Charles V. Riley, Alpheus S. Packard, and Cyrus Thomas, *First Annual Report of the United States Entomological Commission for the year 1877 Relating to the Rocky Mountain Locust* (Washington, D.C.: Government Printing Office, 1878), 53–54, 109, 113. An account of the damage caused by the Rocky Mountain locust in Minnesota during the 1870s is found in Annette Atkins, *Harvest of Grief: Grasshopper Plagues and Public Assistance in Minnesota, 1873–1878* (St. Paul: Minnesota Historical Society Press, 1984).

[2] Charles V. Riley, *Seventh Annual Report on the Noxious, Beneficial and Other Insects of Missouri* (Jefferson City: Regan & Carter, 1875), 135.

[3] Peggy Engledow, "Grasshoppers in the Deed Books," *Plum Creek Almanac*, 17 (1999), 76. This information is also found in Riley, et al., *First Annual Report of the United States Entomological Commission*, 73.

of the Rocky Mountain locust, and it was referred to then, as it would be for years to come, simply as a grasshopper. The notation that the grasshoppers "came" and "remained all fall" is presumptive evidence that they were the Rocky Mountain locust rather than a local species of grasshopper with which the residents of Caldwell County would have been familiar.

The invasion of 1847 was only a minor event compared to the following year. "Grasshoppers" began to appear around Austin during the latter part of September 1848 and by the second week in October had increased to the extent that they would "rise up in perfect clouds before the traveler . . . in some instances almost obstructing the progress of the horse." Reports from distant locations confirmed that grasshoppers were found as far north as present-day Waco and as far south as San Antonio. From whence these mysterious insects came and what their final destination might be were matters of speculation. Gardens were destroyed around Austin, but there was no mention of damage to crops, suggesting that most swarms descended on the uncultivated prairies. The locusts remained only a short period of time and, after laying their eggs, suddenly departed.[4]

The locust invasion of 1848 might have been little more than a source of passing interest had not a grim reminder of their visit become manifest during the spring of 1849. The eggs laid during October and November of 1848 began to hatch in early March the following year. Observers noted that the young far exceeded the numbers of adults seen the previous fall. Crawling slowly across the countryside, the schools of young locusts consumed entire fields of corn, wheat, rye, and oats. Gardens were laid to waste within "a few hours." No method of controlling the insects was known, and fear that their crops would be destroyed caused many farmers to delay planting. This strategy proved successful since the young locusts molted into winged adults and began to depart during April and early May while there was still time to put in a late crop.[5]

Farmers quickly realized that birds were their allies in the struggle against the locusts. "Prairie fowls" were observed to eat large numbers of the young insects, thus somewhat mitigating their ravages. To further this form of natural control, it was suggested that this assemblage of birds be protected from February through September.[6] Unfortunately, this recommendation was not heeded, and birds did not receive any statewide protection for many years.

The destruction of crops during the spring of 1849 convinced many Texans that a pest with the potential to "become a serious evil" confronted

[4] "Grasshoppers," *Texas Democrat* [Austin], Oct. 11, 1848 (quotation); Samuel J. Wood, "The Grasshoppers of Texas," *Texas Almanac for 1861* (Galveston: Galveston Weekly News, 1861), 138.

[5] "Grasshoppers or Locusts," *Texas Democrat* [Austin], Mar. 31, 1849 (quotation); *Democratic Telegraph and Texas Register* [Houston], May 10, 1849; *Texas Democrat* [Austin], Apr. 21, 28, 1849.

[6] "Grasshoppers," *Texas Democrat* [Austin], Apr. 21, 1849.

them. The offending insects were correctly identified by the news media as "locusts" [migratory grasshoppers] but were thought to be the same species renowned for causing the plagues described in the Old Testament.[7] This mistaken belief would persist even after the Rocky Mountain locust was described as a new species in 1866 and found to be a permanent resident of the Rocky Mountains.

Farmers were undoubtedly apprehensive after the experience of 1848–1849 that the locusts might return the following year. However, locusts were not seen again until September 1853, when an enormous flight suddenly appeared over Ellis County. Four decades later, the Reverend R. M. White recalled that the approaching swarm appeared as a reddish-tinged "dark cloud" that gradually arose from the horizon. As the swarm drew near, the sound of beating wings resembled a "heavy wind." At first, only a few insects descended, but this number rapidly increased until their bodies covered the ground a couple of inches deep in some places. Even with large numbers on the ground the sun still "looked as if it [were] enveloped in a dense smoke." Within three days all green vegetation was destroyed and the countryside "looked as if a fire had swept over it." The ravenous insects invaded houses and cut holes in clothing as they searched for food. Hogs and domestic fowls grew fat eating the locusts. The "strong, peculiar odor" of the insects tainted the fowls, however, and most people found their flesh to be unpalatable. Most of the locusts left after about two weeks but not before depositing their eggs. The young that hatched during 1854 destroyed much of the native vegetation and did some injury to the wheat.[8] Although locusts undoubtedly invaded other parts of Texas during the fall of 1853, no records other than their occurrence in Ellis County have been found.

Locusts again returned to Texas during the fall of 1854. For three days during October they passed over Fort Chadbourne in Coke County. In mid-November 1854 large swarms appeared around San Antonio. Some people believed their appearance was indicative of disease, although none was reported. Enormous flights were seen coursing in a southerly direction over San Antonio for nearly two weeks. Where they had come from and where they were going was a topic of conversation as people marveled at the glistening hordes flying overhead. One reporter speculated that if they had all fallen to the ground, they would have covered it to a depth of

[7] *Texas Democrat* [Austin], Mar. 28, Apr. 21, 1849.

[8] The recollections of R. M. White are found in *A Memorial and Biographical History of Ellis County, Texas* (Chicago: Lewis Publishing Co., 1892), 98–99. The "strong, peculiar odor" of the locusts White described may have been due to the presence in their tissues of *caloptine*, a "transparent, reddish-brown oil of very pungent and penetrating odor" (Riley, et al., *First Annual Report of the United States Entomological Commission*, 442). Cliff D. Cates, *Pioneer History of Wise County* (Decatur: Cliff D. Cates, 1907), 92, reported that chickens in Wise County died from "crop-swelling" following the ingestion of locusts, which is questionable. The ingestion of locusts is not known to cause the death of chickens or any other type of bird.

eleven feet! The locusts began to disappear by the first week in December but not before depositing multitudes of eggs that would hatch during the coming spring.[9]

A cold spell during late February 1855 raised hopes that the eggs would be killed. However, this did not happen; by early April the young locusts were doing serious damage in Medina County and at San Antonio and Gonzales. Once again the plovers that feasted on the locusts on the prairies around Gonzales demonstrated the potential of birds as natural control agents.[10]

The pattern seen in 1854–1855 was repeated in the following year. The first swarms of 1855 appeared during late September in the vicinity of Fort Belknap in Young County. The locusts totally destroyed the vegetables in many places but apparently did little damage to the other crops. Swarms so extensive that they partially obscured the sun were reported at San Antonio during November. Many locusts fell to the ground and deposited eggs before continuing their southerly flight. Locusts were seen over Longpoint in Washington County for three consecutive days in November. However, the few insects that descended during the night did little damage and laid no eggs. Reports of damage by the young locusts during the spring of 1856 were received from Seguin, Austin, and Tarrant County.[11]

Huge swarms of locusts arrived during October 1856 in Travis County, where they destroyed the young wheat and deposited their eggs before departing. The behavior of the young that hatched during the spring of 1857 was ominous. They were at first sedentary and did little damage to the vegetation. But, when about half grown they "started on foot towards the north, preserving as much regularity in their march as an army of well-drilled soldiers." During their northward trek, they became "perfectly ravenous, destroying almost every kind of tender vegetation." The advancing hordes were stopped by nothing, not even human habitations through which they marched "with impunity." Observers were appalled to find that the juveniles were cannibalistic and would instantly eat those of their own that were disabled. Then, when the juveniles underwent their final molt and acquired wings at about the age of six weeks, they "all arose at once, as if by common consent, and took their flight toward the north."[12] Although

[9] W. P. Johnston, *The Life of General Albert Sydney Johnston, Embracing His Services in the Armies of the United States, the Republic of Texas, and the Confederate States* (Austin: State House Press, 1997), 174; *Alamo Star* [San Antonio], Nov. 13, 1854; *Western Texan* [San Antonio], Nov. 23, 1854; *Galveston Daily News*, Apr. 3, 1855 (exaggerated appraisal that grasshoppers would stack up eleven feet deep if they landed); *Alamo Star* [San Antonio], Dec. 4, 1854.

[10] *Galveston Weekly News*, Mar. 13, 27, Apr. 3, 10, May 8, 1855; "Gonzales," *ibid.*, Apr. 10, 1855.

[11] *Texas State Gazette* [Austin], Nov. 17, 1855; *San Antonio Herald*, Nov. 6, 8, 1855; Jerry B. Lincecum, E. H. Phillips, and P. A. Redshaw, *Science on the Texas Frontier. Observations of Dr. Gideon Lincecum* (College Station: Texas A&M University Press, 1997), 72; *Galveston Weekly News*, May 13, 1856.

[12] Wood, "The Grasshoppers of Texas," 138. An article in the *Texas State Times* [Austin], Apr. 26, 1856, correctly identifies the invaders as locusts rather than the common grasshopper, which does not have the power of continuous flight and is not migratory in its habits.

the damage was extensive in Travis County, most of Texas was spared during the invasion of 1856–1857.

In late October 1857, the locusts returned with a vengeance. Swarms first appeared in Laredo, moving in a southerly direction taking a swath about seventy miles in width. Orange and lemon trees were destroyed forty miles upriver from Brownsville. Later reports placed swarms at Austin, San Antonio, Gonzales, San Marcos, Seguin, and La Grange, where they devoured gardens and fields of wheat and rye. The editor of the *Texas Sentinel* described graphically the mesmerizing quality of the airborne hosts at Austin:

> By shading the disc of the sun and peering into the flood of golden light just above it, the eye beheld what [might be mistaken] for a shower of shooting meteors. . . . [T]heir apparently white silvery bodies [which] glistened in the noontide rays of the sun, presented a glorious spectacle. Some hovered near the surface of the earth, where their wings and forms were distinctly delineated. Others, venturing upon a bolder and loftier flight, resembled stars, shooting from their spheres and madly rushing through the fields of air.[13]

Depredations of the locusts around Gonzales and Seguin compounded what was already a serious condition. Hard freezes during the spring of 1857 followed by summer drought had resulted in a complete failure of the crops. The locusts then destroyed the fall gardens. Around the end of November the locusts began to depart, but, as one observer gloomily noted, they had remained long enough to "preclude the possibility of any crops, next year, except a crop of themselves." The locusts had in fact deposited enormous numbers of eggs. At La Grange, one observer counted one hundred holes within a square foot, each of which contained a pod with six or seven eggs. Studies by the Entomological Commission later found that individual females could lay an average of three pods of eggs, each of which might consist of as many as thirty eggs.[14]

Some of the eggs began to hatch in late January 1858, raising farmers' hopes that the juveniles would mature and be gone before the spring crops were planted. This hope, however, was soon abandoned. Eggs were still hatching during late March, and the young covered the prairies from Goliad west to the Nueces River. Later reports placed them at Austin, Gonzales, Hallettsville, Laredo, Lockhart, San Antonio, San Marcos, and Webberville. Hoping to gather specific information that would help farmers control the

[13] "Texas Affairs," *Galveston News*, Oct. 29, 1857; *Texas Sentinel* [Austin], Jan. 2, 1858; *Galveston Tri-Weekly News*, Nov. 17, 24, Dec. 3, 1857; "Texas Affairs," *Texas Sentinel* [Austin], Nov. 28, 1857 (quotation); *Galveston Weekly News*, Dec. 1, 8, 1857.

[14] *Galveston Tri-Weekly News*, Nov. 17, 1857. Details of climatic conditions during 1857 can be found in S. D. Casto, "The Texas Gray Squirrel Migration of 1857," *East Texas Historical Journal*, 41, no. 1 (2003), 48–50; "Texas Affairs: San Marcos," *Galveston Weekly News*, Dec. 8, 1857 (quotation); "La Grange, Fayette Co.," *Galveston Weekly News*, Jan. 12, 1858; Riley, et al., *First Annual Report of the United States Entomological Commission*, 228.

locusts, the editor of the *Texas Sentinel* asked readers to answer a series of questions and send their reply to the paper:

> When the insects arrived at any given point—their course of flight—how wide their column, both when in flight and when feeding—what devastations they committed—what kind of vegetation did they first attack—when and how did they deposit their eggs—when were the young hatched out in the spring—what vegetation did the young first attack—how long were they in maturing—when they took their first flight, their course, and other such information . . . as might prove useful in the future.[15]

Regrettably, this request for information did not receive much of a response. The farmers of Texas were busy just trying to survive and had little time to devote their attention to things of a scientific nature.

The counties most affected by the locusts during the spring of 1858 included Bexar, Caldwell, Comal, DeWitt, Goliad, Gonzales, Guadalupe, Hays, Karnes, Lavaca, Nueces, and parts of Bastrop, Fayette, and Victoria. The insects ravenously devoured corn, wheat, cotton, and vegetables and seriously damaged fruit trees, shrubbery, and shade trees. The only agricultural crop that seemed to escape the attention of the locusts was the sorgo or Chinese sugarcane (*Sorghum saccaratum*).[16]

The advance of the juvenile locusts was gallantly resisted. In Austin, residents used "boxes, barrels, buckets, quilts, [and] sheets" to cover plants, with the result that the gardens of the city looked like "tented fields." Entire families "armed with brushes, brooms, and other formidable weapons, rushed forward to do battle. . . . Huge smokes were raised [and] droves of hungry chickens hurried to the scene of action." Finally, wearied from a struggle that could not be won, the defenders retreated and the locusts were declared the victors. Farmers around Austin were advised to keep their corn and cotton fields free of grass in the hope that this would provide some degree of protection from the advancing armies of locusts. A farmer at Gonzales reportedly caught a hundred pounds of locusts in three hours while another gentleman, perhaps with some exaggeration, claimed to have removed four thousand pounds of them from his garden during a single week.[17]

The juvenile locusts underwent their final molt and acquired wings during late April and early May. For several hours huge flights were seen passing over San Antonio from "forty or fifty feet high as far as the eye could reach." In Bastrop "countless millions" were seen flying overhead.

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[15] *Galveston Weekly News*, Jan. 26, 1858; "Texas Items," *Galveston Weekly News*, Mar. 23, 1858; "The Grass-hoppers," *Texas Sentinel* [Austin], Apr. 14, 1858 (quotation).

[16] *Weekly Independent* [Belton], May 15, 1858; *San Antonio Daily Herald*, Apr. 27, 1858; *Weekly Independent* [Belton], May 15, 1858.

[17] *Texas Sentinel* [Austin], Apr. 17, 1858 (quotations); *Southern Intelligencer* [Austin], Apr. 14, 1858; *Texas Sentinel* [Austin], Apr. 24, 1858.

In Austin, Blanco, and Gatesville the migrating swarms descended briefly and did considerable damage before continuing their northward journey. A general rain fell during the first week of May and, with the locusts departing, the farmers began to replant the crops that had been earlier destroyed. This effort was successful, and it was later reported that the crops of 1858 were some of the best seen for the previous three years.[18]

The fall of 1858 was a time of apprehension for many farmers. Some fields along the Colorado River below Austin were left unplanted because of fear that the locusts would return. And, indeed, the first indications were that they would. Hordes of locusts north of the Red River were reported heading for Texas. Large swarms soon arrived in Parker and other north central counties. Locusts were seen flying southward high over Austin, but they did not alight. Countless myriads passed over the city of Belton during early October. In mid-October large flights were seen flying westward over Hamilton. Flights occurred over Belton during the last week in October, and many descended onto the surrounding prairies and fields but did little damage except to the turnips and an occasional field of wheat. Swarms flying southward were reported at several locations during the first two weeks of November. Where these swarms finally descended is unknown, but it was apparently not in Texas. The only report of damage by the immature locusts during the spring of 1859 was in Belton, where the adults had laid their eggs during the previous October.[19]

Texas was relatively unaffected by the Rocky Mountain locust during the 1860s. There were no visits during the Civil War. Then, in mid-October 1866, the locusts suddenly appeared north of Dallas, where they did extensive damage to wheat in Collin and Grayson Counties. Swarms were later reported in McLennan County and as far south as Atascosa and Medina Counties. The eggs laid during the fall of 1866 began to hatch during early March 1867. The emerging juveniles did extensive damage to the wheat in Bell, Coryell, and Burnet Counties before their numbers were reduced by a severe freeze that occurred on March 13.[20] This freeze was perhaps a key factor in preventing damage in other parts of the state.

The locusts returned again during mid-October 1867. Enormous swarms were seen west of Dallas. In Bosque County they devoured all of the turnips

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[18] *San Antonio Daily Herald,* Apr. 30, 1858 (1st quotation); *Galveston Weekly News,* May 25, 1858 (2nd quotation); *Galveston Weekly News,* May 11, 25, June 8, 1858; "Crops in Western Texas," *San Antonio Daily Herald,* May 22, 1858; *Southern Intelligencer* [Austin], May 19, 1858; *Weekly Independent* [Belton], May 22, 1858.

[19] "Letter from Austin," *San Antonio Daily Herald,* Oct. 27, 1858; "Grasshoppers North of Red River," *Dallas Herald,* Oct. 20, 1858; "Texas Items: Parker," *Galveston Weekly News,* Nov. 2, 1858; Wood, "Grasshoppers of Texas," 138; *Weekly Independent* [Belton], Oct. 3, 1858; *Southern Intelligencer* [Austin], Oct. 20, 1858; *Belton Independent,* Oct. 30, 1858; *Galveston Weekly News,* Nov. 16, 1858; *Belton Independent,* Mar. 19, 1859.

[20] *Galveston Tri-Weekly News,* Oct. 29, 1866; *Galveston Tri-Weekly News,* Nov. 7, 9, 1866; Lincecum, et al., *Science on the Texas Frontier,* 73.

and other garden vegetables and stripped peach trees bare. The swarms in DeWitt County laid millions of eggs while at Gonzales many of the recently laid eggs were already hatching. Additional regions infested by the adult locusts included Austin, Bell, Burleson, Coryell, Fannin, Fayette, Hunt, Lampasas, Lavaca, Red River, and Washington Counties.[21]

Gideon Lincecum, a physician and naturalist living in the community of Longpoint, made detailed observations on the locusts that appeared in Washington County during November 1867. Lincecum was amazed at the swarming behavior of the locusts, but it was their reproductive activities that most fascinated him. Over a period of days, he observed the females as they selected the ground in which to deposit their eggs, as well as their method of constructing the holes into which the egg pods would be deposited. Lincecum took measurements of the pods and counted the number of eggs contained in each pod. The locusts continued to couple following a cold spell in late November even though large numbers of them were already dead or dying. Cannibalism was increasingly observed as vegetation became scarce. The remaining locusts finally took wing and departed in a southerly direction during the first week in December.

Lincecum appears to be the only naturalist to report a change in body color of the Rocky Mountain locust following its death. He first noted that the locusts turned red when placed in alcohol. Individuals dying in the field of natural causes also turned red after a few hours. Even more surprising, Lincecum found that excrement of the chickens that ate the insects had a distinct reddish color. Based on these observations, Lincecum speculated that the locusts contained a substance that might be of value as a dye.[22]

The young that hatched during the spring of 1868 did considerable damage to wheat and corn in Collin, Denton, and Navarro Counties. A farmer at Kaufman reported that locusts had destroyed his entire field of corn and in some places were gathered "in bunches as big as a flour barrel." Thomas Affleck, a horticulturist at Brenham, was assisted in his struggle to save his orchard and garden by a large flock of blackbirds. Later, when the schools of young locusts began to depart from his property, they were

[21] "Texas Items: The Grasshoppers are Coming," *Galveston Daily News*, Oct. 23, 1867; *Galveston Daily News*, Nov. 15, 16, 1867; B. D. Walsh, "The Hateful Grasshopper (*Caloptenus spretus* Walsh)," *First Annual Report of the Noxious Insects of the State of Illinois*, appendix of *Transactions of the Illinois State Horticultural Society* (Chicago: Prairie Farmer Company Steam Print, 1868), 132–133; Riley, et al., *First Annual Report of the United States Entomological Commission*, 58.

[22] The observations made by Lincecum during November and December 1867 are found in his diary and in letters written to Joseph Leidy at the University of Pennsylvania and Spencer Baird at the Smithsonian Institution. Lincecum, et al., *Science on the Texas Frontier*, 71–76; "Diary and Papers of Gideon Lincecum, 1846–1870," Box 2R79, Gideon Lincecum Collection (Center for American History, University of Texas at Austin).

followed and preyed upon by hundreds of Harlan's red-tailed hawks (*Buteo jamaicensis harlani*).[23]

The passage of locusts on two successive days over Sherman during mid-October 1868 raised fears that a major invasion had begun. However, none of the insects alighted. A second report of multitudes of grasshoppers at Denton during mid-November represented the only other sighting during the fall of 1868.[24] No record has been found of damage done by the immature locusts during the spring of 1869.

The origin of the swarms of locusts was first made known to Texans in the December 1868 issue of *The Plow Boy* (Austin). An article reprinted from *American Entomologist* revealed that the native home of the migratory insect called the hateful or Colorado grasshopper was in the canyons of the Rocky Mountains.[25] The article also presented information on the behavior of the swarms during their migration, but in 1868 virtually nothing was known of the natural history of the species in its permanent home in the mountains of western America and Canada.

The first locusts of the 1870s appeared in Stephenville during early October 1872. For an entire day they passed overhead but in the evening settled to the earth in countless millions. They continued to pass overhead the following day with additional multitudes alighting on the ground. Farmers immediately stopped planting wheat with the plan that it would not be sown until the insects departed. Swarms arrived in Fredericksburg during mid-October, and it was feared that they would soon reach San Antonio. Later reports placed them in Gonzales, Lampasas, Medina, Williamson, and Wilson Counties.[26]

The damage done during the fall of 1872 was apparently of little consequence, and it was not until the following spring that the full impact of the invasion was known. The wheat crop around Stephenville was "nearly destroyed" by juvenile locusts. A forty-acre field of wheat in Lampasas County was destroyed in four days. Two children in Medina County became ill after drinking from a well into which a great number of "grasshoppers" had fallen. Enormous flocks of sparrowlike birds (probably dickcissels) suddenly appeared in Gonzales and, much to the relief of the farmers, began to eat the locusts.[27]

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[23] *Dallas Weekly Herald*, Apr. 25, 1868; *Galveston Daily News*, May 6, 1868; *Dallas Herald*, May 16, 1868; *Galveston Daily News*, May 5, 1868 (quotation). The observations of Thomas Affleck at Brenham are reported in Riley, *Seventh Annual Report of the Noxious, Beneficial, and Other Insects of the State of Missouri*, 191–193 and Riley, et al., *First Annual Report of the United States Entomological Commission*, 58.

[24] *Galveston Daily News*, Oct. 27, 1868; *Weekly Harrison Flag*, Nov. 26, 1868.

[25] "The Hateful or Colorado Grasshopper," *Plow Boy* [Austin], Dec. 1868.

[26] "Grasshoppers," *Galveston Daily News*, Oct. 11, 1872; "State News," ibid., Oct. 20, 1872; *Fayette County New Era*, Nov. 14, 21, 1872.

[27] *Galveston Weekly News*, May 5, 1873 (1st quotation); *North Texas Enterprise* [Bonham], May 10, June 7, 1873 (2nd quotation); *Galveston Daily News*, May 16, 1873.

The locusts appeared again during the fall of 1873. The first flights were seen in Hood County during early October. Swarms passed over Comanche County but did not alight, whereas in Johnson County the insects descended and helped the farmers "pick their cotton." Cotton and wheat were damaged in Parker and other western counties. Around Dallas the damage was localized, and the locusts soon left the area. Some locusts remained in Collin County until at least mid-November while in McLennan County many passed over but did not descend.[28] There seem to be no reports of serious damage occurring during the spring of 1874.

The first swarms in the fall of 1874 appeared during mid-August at Gainesville, where they seriously damaged cotton. Immense swarms were next observed flying over Eagle Pass in Maverick County. These swarms suddenly appeared during September and for five consecutive days flew over the city, alighting at night and continuing their migration southward along the Rio Grande on the following morning. The adult locusts did great damage during their nocturnal grounding, as well as laying immense numbers of eggs that began to hatch during January the following year. The eggs were laid in a two-mile swath with the Rio Grande at its center and extending south for an undetermined distance. Crops were particularly damaged during the fall of 1874 in the north central counties of Cooke, Young, and Palo Pinto, and the south central counties of San Saba, Gillespie, Blanco, Kendall, Bandera, Medina, and DeWitt.[29]

The damage done by immature locusts during the spring of 1875 seems to have been extensive in some areas. In "Onion Valley" in Ellis County "not a spear of grass or anything green was to be seen [within] a radius of twenty-miles." Other regions reported little damage. During mid-May the locusts around Denton began to die, and "bushels" of them were found in places. Each dead locust was found to contain one to three white "worms." Later investigations suggested that these worms were probably the larvae of tachinid flies. Huge flocks of migrating longbilled curlew, golden plover, and upland sandpiper cleared the fields of young locusts in Dallas County and were credited with having saved many of the crops in that area.[30]

Locusts visited only a few locations during the fall of 1875. At Eagle Pass the swarm that arrived during September was judged to be less "formidable"

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[28] *Denison News*, Oct. 10, 1873; *Houston Daily Mercury*, Oct. 14, 1873 (quotation); *North Texas Enterprise* [Bonham], Oct. 25, 1873; *Dallas Weekly News*, Nov. 1, 1873; *Houston Daily Mercury*, Oct. 26, Nov. 6, 1873.

[29] *North Texas Enterprise* [Bonham], Aug. 21, 1874; Riley, et al., *First Annual Report of the United States Entomological Commission*, 60; Riley, *Seventh Annual Report of the Noxious, Beneficial and Other Insects of the State of Missouri*, 144. Riley reported locusts doing significant damage in "Belknap" County. Since there is no Belknap County in Texas, he must have been referring to the region of Fort Belknap in Young County.

[30] *Dallas Weekly Herald*, May 22, 1875 (1st quotation); *Denison Daily Cresset*, May 12, 1875 (2nd quotation); Riley, et al., *First Annual Report of the United States Entomological Commission*, Appendix III, 66.

than the one that had visited the previous year. Small numbers of locusts were also seen at Laredo and Uvalde. "Clouds" of grasshoppers were seen passing over Denton during early October.[31] The damage done by these swarms and their progeny that hatched during the spring of 1876 was insignificant.

The immensity of the swarms that descended upon Texas during the fall of 1876 was truly astonishing. In Collin County they came down "so deep as to cover fences and the bodies of trees upon which they would alight." The swarm passing over Dallas was "estimated to be 2,000 feet high, and from forty to sixty miles wide." Locusts were so abundant at Flatonia that their crushed bodies greased the rails, preventing passage of the eastbound train for several hours. Similar train stoppages occurred in Austin, San Antonio, and on the tracks near Marion in Guadalupe County. A scarcity of good drinking water was reported at Lockhart because the decomposing bodies of the locusts had tainted most of the local wells.[32]

The first swarms of 1876 appeared during mid-September in Grayson, Jack, Palo Pinto, and Wise Counties, and by the end of the month they had reached San Antonio. Many farmers believed that the insects had arrived too late to damage cotton and too early to hurt wheat that had not yet been planted. In Hill County some farmers suggested that the locusts might even "do some good by eating the broom weed." Later reports were not so optimistic. Every shrub and vegetable, and all nursery stock, were stripped bare of foliage in Dallas. In Navarro County, "all the garden vegetables" were eaten, and young fruit trees were left "entirely bare of leaves and bark." Cabbages, cucumbers, tomatoes, and sweet potatoes were destroyed in Victoria. Large fires were set at night in the vicinity of Calvert in the erroneous belief that the flying insects would be lured to their death in the flames. Most farmers prudently delayed planting wheat until after the locusts departed in early December. Cattle in Bastrop County reportedly died from eating locusts, and stockmen in some areas were concerned that decimation of the prairie grasses, coupled with drought, would result in inadequate forage for livestock during the approaching winter. Although the area infested by the locusts was estimated to be "200 miles in width by 360 in length," the damage they inflicted was localized and relatively insignificant.[33] However, knowing that the swarms had laid untold numbers of eggs raised fears of the destruction that could potentially occur when the young locusts emerged during the coming spring.

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[31] Riley, et al., *First Annual Report of the United States Entomological Commission*, Appendix III, 60–61 (1st quotation); *Marshall Tri-Weekly Herald*, Oct. 23, 1875 (2nd quotation).

[32] *Galveston Daily News*, Oct. 4 (1st quotation); 11, 25, 1876; Riley, et al., *First Annual Report of the United States Entomological* Commission, 61 (2nd quotation); *Dallas Weekly News*, Oct. 7, 1876; *San Antonio Daily News*, Oct. 21, 24, 1876.

[33] *Galveston Daily News*, Sept. 21, 22, 23, 29 (1st quotation), Oct. 1, 6 (2nd quotation), 26, 1876; "Cotton Destroyed by Grasshoppers," *Frontier Echo* [Jacksboro], Sept. 22, 1876; Ammon Burr, "Notes from Dallas," *Gardener's Monthly*, 19 (1877), 309–310; "From Clinton," *Victoria Advocate*, Oct. 26, 1876; Riley, et al., *First Annual Report of the United States Entomological Commission*, 61 (3rd quotation), 404.

The winter of 1876–1877 was relatively mild, and the eggs began to hatch during late January and continued hatching through March. The entire surface of the earth at Flatonia was said to be "alive with hoppers," and it was feared that the crops would be destroyed. Corn, wheat, and vegetables were the most seriously affected. Many crops were replanted one to three times in Limestone and other counties. At Denison the blooms on 12,000 strawberries were destroyed. In Ellis County the young locusts ate the leaves off the bushes in the creek bottoms "to the height of a man on horseback" and were occasionally seen clustered in trees "twenty feet above the ground." However, in most places, the widespread damage that had been feared did not occur. The numbers of young locusts at San Antonio were so reduced by cold, drenching rain and birds that it was predicted they would do little damage. Prairie fires in Jack County killed countless millions, while in other areas large numbers of the insects died after being parasitized by tachinid flies. In a few places the crops were entirely destroyed, but in Dallas the yield was anticipated to be the best "in several years." Indeed, the prospects for farming were so good that the *Dallas Herald* proclaimed that persons wanting to move to Texas "need not fear to come . . . on account of grasshoppers." Some farmers actually considered the locusts to be a "disguised blessing" since they destroyed the broomweed (*Gutierrezia* spp.) and other pestilential species of plants. It was estimated that 5 percent of grain crops and 25 percent of garden vegetables were destroyed in Texas during the spring of 1877.[34]

The damage caused by locusts during the mid-1870s was so extensive that in March 1877 Congress authorized the formation of the United States Entomological Commission "to study and make recommendations for control of the Rocky Mountain locust in the Western States and territories." Three eminent entomologists were appointed to carry out the mission: Charles Valentine Riley, Alpheus Spring Packard Jr., and Cyrus Thomas. Riley, because of his previous studies of the Rocky Mountain locust, was appointed chief of the commission. The geographic area to be studied was divided among the three members with Riley choosing to study the locust in the southern regions of its distribution. He first came to Texas during April, later visiting Missouri, Kansas, Iowa, Nebraska, Colorado, and Canada before again returning to Texas in November.[35]

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[34] Riley, et al., *First Annual Report of the United States Entomological Commission*, 61, Appendix III, 82; *Galveston Weekly News*, Mar. 5, 1877 (1st quotation); *Frontier Echo* [Jacksboro], May 4, 1877; W. H. Getzendaner, *Weather Memoranda Extremes, Ellis County, Texas* (Waxahachie: Ellis County Mirror, 1909), 3 (2nd quotation); "Letter from San Antonio," *Galveston Weekly News*, Mar. 5, 1877; "State News from All Sections: Jack County," *Galveston Weekly News*, Apr. 9, 1877; "The Grasshoppers," *Dallas Weekly Herald*, Apr. 28, 1877 (3rd quotation); "Leaving the State," *Dallas Herald*, May 3, 1877; Riley, et al., *First Annual Report of the United States Entomological Commission*, Appendix III, 70 (4th quotation), 71 (5th quotation); Charles V. Riley, A. S. Packard, and Cyrus Thomas, *Second Report of the United States Entomological Commission for the Years 1878 and 1879* (Washington, D.C.: Government Printing Office, 1880), Appendix V, 57–61.

[35] R. W. Dexter, "The Organization and Work of the U.S. Entomological Commission (1877–1882)," *Melsheimer Entomological Series*, 26 (1979), 28–32 (quotation); Riley, et al., *First Annual Report of the United States Entomological Commission*, letter of transmittal, p. xi.

C. V. Riley (1843–1895) was born in London, England, and came to the United States when he was seventeen years old. He was first employed as an editor in the entomology department of the *Prairie Farmer* and later as the state entomologist of Missouri. His numerous publications in the latter position earned him wide acclaim as the foremost economic entomologist of his time. Riley was also the cofounder and editor of the *American Entomologist*, a leading journal of its time. Riley was not only a prolific author, he was also an outstanding artist who illustrated his own publications.[36]

Riley soon realized that he would need an assistant in Texas, and Jacob Boll (1828–1880), a naturalist then living in Dallas, was chosen for the job. Boll was a "gifted and versatile student of both botany and zoology" and was well known by other naturalists of his day. In 1869, he left his native Switzerland for Texas, where he collected for Louis Agassiz of Harvard for about a year. He later returned on two occasions to Switzerland, but, following the death of his wife in 1873, returned to Dallas, where he remained until his death in 1880. Boll collected insects for museums all over the world, and it was perhaps his knowledge of entomology that persuaded Riley to hire him as a "special assistant."[37]

The extent of Boll's contribution to the locust study is unknown. His observations in Bexar, Comal, Dallas, Harris, Limestone, Tarrant, and Williamson Counties are included in Appendix III of the first annual report of the entomological commission. Letters that he wrote to the Dallas newspapers also contain some of his observations on the natural history of the locusts. After observing tachinid flies parasitizing locusts around Dallas, Boll predicted that the parasite would soon minimize their numbers. In a second letter, he noted that the destruction of the spring buds of broomweed by the locusts would greatly reduce the numbers of this pest weed for years to come. Boll may have also been involved in the collection and shipment of locusts to Riley's headquarters in St. Louis for experiments on the reproduction of the insects. Although Boll collected extensively in Texas, only a single specimen of the Rocky Mountain locust—a male he collected in Dallas—exists today in museum collections. This specimen, now in the Smithsonian National Museum of Natural History, is the only known physical documentation of the invasion of Texas during 1876–1877.[38]

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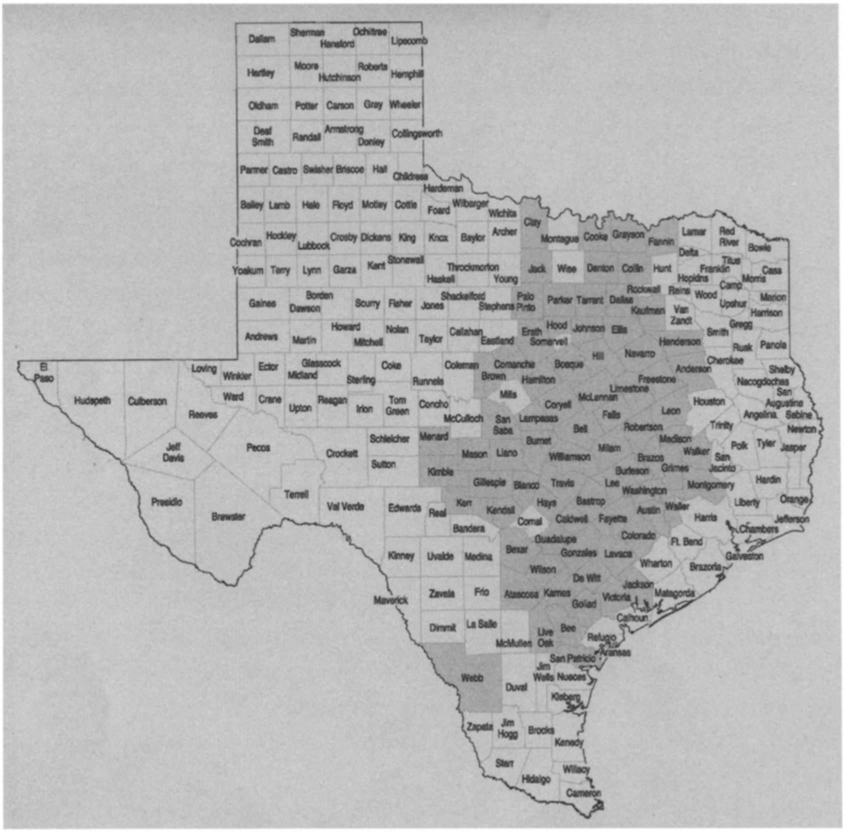
[36] Dexter, "Organization and Work of the U.S. Entomological Commission," 28–32.

[37] S. W. Gesier, "Professor Jacob Boll and the Natural History of the Southwest," *American Midland Naturalist*, 11 (1929), 435–452.

[38] *Dallas Weekly Herald*, Apr. 28, 1877; letter reprinted in Riley, et al., *First Annual Report of the United States Entomological Commission*, Appendix III, 72. Two additional letters by Boll describing the arrival of locusts at Dallas in September 1876 were published in *Neue Zurcher Zeitung* [Zurich], Nov. 1 and 2, 1876. Abstracts of these two letters are found in A. S. Packard, "Report on the Rocky Mountain Locust and Other Insects Now Injuring or Likely to Injure Field and Garden Crops in the Western States and Territories," in F. V. Hayden, *Ninth Report of the United States Geological Survey of the Territories* (Washington, D.C.: Government Printing Office, 1877), 609–610. Riley, et al., *First Annual Report of the United States Entomological Commission*, 277; A. B. Gurney and A. R. Brooks, "Grasshoppers of the Mexicanus Group, Genus *Melanoplus* (Orthoptera: Acrididae)," *Proceedings U.S. National Museum*, 110 (1959), 1–93.



Texas counties reporting the Rocky Mountain Locust, September 1876–May 1877. *An original map prepared from reports published in various media during the years 1876–1877.*

Riley preferred the personal touch when dealing with the media. During late April, he visited the offices of the *Dallas Weekly Herald* to discuss the locust problem. The result of this meeting was a feature article in which Riley assured Texans that their state would never suffer to any great extent from the locusts since they came too late in the fall and left too early in the spring to seriously affect most crops. Although Riley already had a network of "intelligent correspondents," he was still seeking additional information from all regions of the state. To assist in achieving this objective, the *Herald* published a list of twenty-two questions on the natural history and control of the Rocky Mountain locust. Those desiring a copy of a circular containing the same questions would be sent one by mail.[39]

The network of correspondents that Riley assembled in Texas included thirty-five private citizens, twenty-two of whom were from only seven locations:

[39] *Dallas Weekly Herald*, Apr. 28, 1877 (supplement).

Austin (7), Dallas (5), Galveston (2), Denison (2), Bastrop (2), Calvert (2), and Clifton (2). In addition, Riley obtained information from the United States Signal Corps observers at Corsicana, Denison, Eagle Pass, Fredericksburg, Galveston, Laredo, Mason, Pilot Point, and Uvalde. Sgt. C. A. Smith, Signal Corps observer at Galveston, provided yeoman service by extracting and summarizing 282 reports of locusts appearing in the exchange columns of the *Galveston Daily News* from January through May 1877.[40] No voucher specimens were collected by any of Riley's correspondents. This lack of documentation is perhaps due to the fact that most of the young locusts were already in the process of leaving Texas by the time the study began in April 1877.

Riley's work in Texas was completed in November 1877. In February the following year the members of the commission met to discuss their data and prepare their report. This report, published in late 1878, was devoted entirely to the history, classification, biology, and means of control of the Rocky Mountain locust. Included in the report were sections on the history of the locust in Texas, migrations prior to 1877, an extensive appendix giving detailed observations for 1877, as well as various miscellaneous observations. Additional data on the damage done by locusts during the spring of 1877 was later included in the second annual report of the commission, published in 1880.

The fall of 1876 and spring of 1877 marked the last major incursion of the Rocky Mountain locust into Texas. Locusts were not reported during fall 1877, spring 1878, or fall 1878. During May 1879 swarms were seen at Dallas, and in the autumn of that year locusts were reported at Eastland. During the spring of 1880 grasshoppers were reported in Denton and Coryell Counties, and in Weatherford.[41] Although assumed to be Rocky Mountain locusts, these isolated occurrences during the fall of 1879 and the spring of 1880 may have been of some other species.

The last visitation of the Rocky Mountain locust occurred during September and October 1880. Large swarms that nearly "obscured the sun" were seen around Jacksboro. Later reports placed them in Wichita and Tarrant Counties, in Dallas, and west of Corsicana.[42] Although many of the locusts descended and presumably laid eggs, there seem to be no reports of damage by the young during the spring of 1881.

Grasshopper outbreaks that occurred during the early 1880s were probably of native species such as the migratory grasshopper (*Melanoplus sanguinipes*) and the red-legged grasshopper (*Melanoplus femurrubrum*), both

[40] Riley, et al., *First Annual Report of the United States Entomological Commission*, Appendix XXVI, 271, Appendix III, 63–82.

[41] Charles V. Riley, Alphaeus S. Packard, and Cyrus Thomas, *Third Report of the United States Entomological Commission* (Washington, D.C.: Government Printing Office, 1883), 3–4; *Weatherford Exponent*, Mar. 13, 1880; *Belton Journal*, Apr. 15, 1880; *Williamson County Sun*, May 27, 1880.

[42] Riley, et al., *Third Report of the United States Entomological Commission*, 3–4.

of which occasionally gather into large swarms. In early July 1885 reports reached San Antonio that locusts had appeared in western Texas. A few days later, a train on the Texas Central and Mexican Railroad 130 miles south of Eagle Pass encountered a swarm so dense that for "two miles the sun was hidden from view." Specimens procured by the passengers were proclaimed to be those of the "genuine Kansas variety." Locusts also appeared in Colorado during July but upon investigation proved to be of a local species. Disturbing reports were received from Chihuahua, Mexico, during September 1885 that locusts were destroying the crops and that the farmers were attempting to frighten and kill them by firing into the swarms with shotguns. These locusts were believed to have entered Mexico from the United States [most likely Colorado and Texas] but whether they were Rocky Mountain locusts or another migratory species is unknown.[43]

Texans responded to the locust invasions of the nineteenth century with a mixture of astonishment, apprehension, active resistance, resignation, humor, and an abundance of invective. Locusts were described in the media as "migratory marauders," "pestiferous devotees of vegetation," "hopping devils," "harbingers of evil," "pestilential destroyers," "dreaded little visitors," "a miserable carpet-bag tribe," "pestiferous varmints," "an invading army," "the greatest of terrors," and as being villainous. The language seemed almost lacking in words sufficient to describe the numbers of locusts and the destruction they wrought. Swarms were said to be immense, huge, or enormous, whereas the number of individual insects was expressed as myriads, innumerable, countless, multitudes, and millions. The locusts were said to bring waste, calamity, incalculable injury, devastation, sad havoc, desolation, and destruction. Depredations of the locusts were considered a form of robbery, an ill, and a scourge to be suffered by those affected. Indeed, the citizens of Texas felt as if they were besieged by the hordes of locusts and their fearful depredations.[44]

Many Texans accepted the locusts as simply one more burden to be borne without complaint. Some, however, fought back with the only tools at their disposal. The citizens of Austin used children, chickens, blankets, and smoke in their unsuccessful attempt to save their gardens during the spring of 1858. In that same year, some planters spread "domestic" (cotton cloth) over their gardens to repel the locusts only to find to their horror that the insects ate the cloth to get at the vegetables. A resident of Austin proclaimed that dry ashes put on young corn would make it unpalatable to locusts. This method of control was, however, not a reasonable alternative

[43] "From San Antonio," *Galveston Weekly News*, July 16, 1885 (1st quotation); "Clouds of Grasshoppers," *Galveston Daily News*, July 12, 1885; "A Grasshopper Plague," *Fort Worth Gazette*, July 18, 1885 (2nd quotation); "The Locust Plague," *Fort Worth Gazette*, July 20, 1885; "Special Telegram to the Statesman," *Austin Daily Statesman*, Sept. 27, 1885.

[44] These epithets and descriptive terms appeared in Texas newspapers between 1848 and 1878.

for large acreages. Many farmers simply replanted or waited for the locusts to depart before putting in their spring crop. Others placed their faith in the native birds and other enemies of the locusts, as well as adverse climatic events to reduce the numbers of insects to a tolerable level.[45] Plowing during December and early January was found to expose the eggs, causing them to die. Combating the pests was the personal struggle of individual farmers. No assistance was expected from the state, and none was given. In fact, there was none to give. Entomological expertise was lacking, and there was no governmental agency to provide resources for control of the insidious marauder.

Farmers in the midwestern states patented various contraptions during the 1870s designed to crush, bag, burn, or poison locusts. Devices to kill locusts were also developed in Texas. The *Marshall Tri-Weekly Herald* reported in October 1875 that a Texan had invented a machine and discovered a poison that would destroy locusts "acres at a time." Unfortunately, the name of the inventor and the details of his machine and poison were not given. A "grasshopper destroyer" developed by the horticulturist Thomas Volney Munson was used around Denison during the spring of 1877. Another apparatus invented by J. C. Melcher of Fayette County crushed the locusts as it was pushed forward into the schools of wingless juveniles. An Ellis County man reportedly invented a machine in 1878 that would kill a bushel of locusts per hour.[46] The design of these homemade devices is unknown, and there is no evidence that any of them were ever patented. It is likely that they were used only locally and were of limited value in the control of the hordes of locusts.

A method of destroying the schools of young locusts was discovered by accident during the spring of 1877. A resident of Navarro County noticed that the juveniles fell readily into postholes that were being dug. Extrapolating from this observation, he proposed to ditch around his fields, drive the young locusts into the ditches, and then cover them with a large plow.

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[45] *Texas Sentinel* [Austin], Apr. 17, 1858; *Southern Intelligencer* [Austin], Apr. 28, 1858; *Texas Sentinel* [Austin], May 1, 1858. Several species of Texas wild birds were reported to eat the Rocky Mountain locust. Prairie chickens fed on the winged insects in the fall, scratched out and ate the buried eggs during the winter, and feasted on the emerging juveniles during the spring. The huge flocks of migrating plovers and curlews, in conjunction with meadowlarks and blackbirds, were particularly effective in reducing the numbers of young that hatched during the spring. Domestic chickens, turkeys, guineas, ducks, and geese voraciously ate locusts but became satiated within a few days. The only specific mention of guineas eating locusts was in Laredo during 1875–1876. Riley, et al., *First Annual Report of the United States Entomological Commission*, 339, Appendix III, 64–67, 73, 75; Riley, et al., *Second Report of the United States Entomological Commission for the Years 1878 and 1879*, Appendix V, 60–61.

[46] R. Douglas Hurt, "Grasshopper Harvesters on the Great Plains," *Great Plains Journal*, 16 (1977), 123–134. Riley, et al., *First Annual Report of the United States Entomological Commission*, 362–377; *Marshall Tri-Weekly Herald*, Oct. 23, 1875 (1st quotation); *Denison Daily Cresset*, May 11, 1877 (2nd quotation); Riley, et al., *First Annual Report of the United States Entomological Commission*, 371; *Williamson County Sun*, May 9, 1878.

Apparently unknown to this resourceful individual was the fact that ditching was already a proven method of destruction widely used in other states.[47]

Various methods of control were used in Dallas during the spring of 1877. Lime and the insecticide Paris green were tried with little effect. One man modified his cultivator with a netlike cloth, which served as a seine to gather the insects. Children were paid five cents a pound for locusts caught with a hand net in an orchard. Some people smeared pine tar on the trunks of their fruit trees to prevent the locusts from ascending into the foliage. This method was effective except on very young plants, which were killed by the tar. Burning of the prairie and of straw placed in the fields at the time the eggs were hatching also proved beneficial in reducing numbers of the young insects. A farmer in McLennan County found that cornmeal soaked in strychnine killed a high percentage of the juveniles, while a family in Brazos County prevented damage through a liberal use of "fire, boards, carbolic soap, tar, kerosene, and turpentine."[48]

The Rocky Mountain locust was occasionally a topic of humor and satire. It was jokingly said in Comanche County that the locusts were "helping the farmers pick their cotton" and that they were willing "to eat everything on the farms except the mortgages." In San Antonio it was declared that more grasshoppers were killed around the saloons than elsewhere in the city "because there are more tramps around them." Locusts were also seen "loafing" on the plaza in San Antonio "but, as usual, no policeman was about." Passersby in San Antonio were also amused to see a deranged woman attempting "to frighten away the depredators with a tin rattle and queer movements of her body." An Austinite gleefully described a half-frozen locust as "limping [along] with a sprained ankle, and an icicle hanging to the end of his nose." The first swarms to arrive in DeWitt County were said to have retreated because they had "heard of so much killing in the county, they were afraid to pass through." The *Dallas Weekly Herald*, in a pointed criticism of the entomological commission, proclaimed, "The most effective commissioner we have seen yet is the [tachinid] fly which deposits the grub under [the wings of the locusts] with fatal effects."[49] This derisive remark, along with the suggestion that "liberal arrangements" be made for employment of the flies, was published in the same issue of the paper in which the chief commissioner, C. V. Riley, assured Texans that they would never suffer major damage from the Rocky Mountain locust.

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[47] *Galveston Daily News*, Apr. 9, 1877; Riley, et al., *First Annual Report of the United States Entomological Commission*, 378–381.

[48] Ammon Burr, "Notes from Dallas, Texas," *Gardener's Monthly*, 19 (1877), 309–310; Riley, et al., *First Annual Report of the United States Entomological Commission*, 363, Appendix III, 78, 80 (quotation).

[49] *Galveston Daily News*, Oct. 13, 1876 (1st quotation); *San Antonio Daily News*, Nov. 3, 1876 (2nd quotation); *Waco Daily Examiner*, Oct. 6, 1876 (3rd quotation); *San Antonio Daily News*, Oct. 24, 1876 (4th quotation); *Southern Intelligencer* [Austin], Nov. 24, 1858 (5th quotation); *Victoria Advocate*, Oct. 26, 1876 (6th quotation); *Dallas Weekly Herald*, Apr. 28, 1877 (7th quotation).

Activities of the locusts were occasionally considered beneficial. Some farmers found that cotton was easier to pick after the insects had defoliated it. It was also widely acknowledged that locusts did good work in destroying broomweed and other pest plants. The early hatching of the eggs during January and February caused farmers to delay planting, a tactic that often saved their crops from late freezes had they planted at the first sign of warm weather. One farmer claimed that the locusts were the cause of his second-growth crop of oats being better than the first, which had been earlier eaten to the ground. The opinion was expressed on several occasions that there was a greater good to be divined from the scourge of locusts. A resident of San Antonio declared that if the locusts could eat up the "weak-kneed" farmers who relied on their "fears and apprehensions" rather than on good advice and trust in the Lord, they would prove a "blessing in disguise."[50] The locusts were, in the opinion of this individual, the agent that would test the character and cleanse the ranks of the farmers of Texas.

Two misconceptions regarding the Rocky Mountain locust were perpetuated in the news media. Newspaper accounts compared the Rocky Mountain locust with the locusts of biblical times, thus reinforcing the false idea they were one and the same species. As late as 1876, the *San Antonio Daily News* informed its readers that the insects then plaguing the city were "the regular locust, such as we read of in the bible." The destination of the massive swarms that passed over South Texas was also misinterpreted. It was widely believed that the swarms continued southward until they eventually fell to their deaths in the Gulf of Mexico. However, inquiries by the entomological commission determined that swarms had never been found closer than a few miles from the coast and that there was no evidence that any had ever perished in the Gulf.[51] Most likely, the large swarms that passed over South Texas continued their journey into Mexico before descending.

The fate of those swarms that crossed the Rio Grande into Mexico is not mentioned in contemporary Texas newspapers, and the entomological commission did not investigate this subject. The Mexican government apparently took little notice of locust outbreaks until the late 1870s. Their official report, a massive tome of 705 pages, documents these pests during the period from 1879 through 1886.[52] The last major invasion of the

[50] Riley, et al., *First Annual Report of the United States Entomological Commission,* Appendix III, 71, 72, 76, 81; *Galveston Weekly News,* Mar. 5, 1877 (quotation).

[51] *Texas Democrat* [Austin], Mar. 31, Apr. 21, 1849; *Weekly Independent* [Belton], May 22, 1858; *Galveston Daily News,* Oct. 23, 1867, Nov. 18, 1876; *San Antonio Daily News,* Oct. 3, 1876 (quotation); Wood, "Grasshoppers of Texas," 138; "Home News," *West Texas Free Press* [San Marcos], Nov. 11, 1876. As late as 1907, Cliff Cates (*Pioneer History of Wise County,* 91) claimed that the swarms that fell into the Gulf were washed ashore "in great ridges along the beach." Riley, et al., *Second Report of the United States Entomological Commission,* 100–103.

[52] *Coleccion de documentos é informes sobre la langosta que ha invadido la República mexicana en los años de 1879 á 1886* (México: Oficina tip. De la Secretaría de fomento, 1886).

Rocky Mountain locust in Texas occurred in the fall of 1876, and by 1879 the Rocky Mountain locust was in decline throughout its entire range. It is therefore unlikely that the outbreaks that occurred in Mexico during the 1880s were the result of swarms that arrived from Texas.

Most Texans soon forgot about the Rocky Mountain locust and the tribulations that they had suffered from its visitations in the 1870s. Others, however, did not. The entomological commission continued to study and publish reports on the locusts until 1883. Many farmers in the midwestern states lived in dread that the locusts would someday return in their former numbers, but they did not. The last specimens of the Rocky Mountain locust, a male and female, were collected in Manitoba, Canada, in 1902.[53] Incredible as it seems, this insect, which only twenty-five years earlier migrated in numbers so vast as to obscure the sun, seemed to have disappeared from the face of the earth. "Out of sight, out of mind" would seem to be an appropriate cliché for the Rocky Mountain locust, but such has not been the case. The study of this species has recently taken on a new life as entomologists have attempted to unravel the mysteries of its disappearance. What stimulated this locust to migrate? Why did the major migrations cease after 1880? Did the Rocky Mountain locust actually become extinct in the early 1900s, or does it still survive in isolated regions of its original mountain home?

There is now a plausible explanation of what caused the Rocky Mountain locust to migrate. The species is believed to have existed in two forms—a solitary form and a migratory form. The solitary form lived in the mountain valleys and along rivers and streams in the Rocky Mountains. When populations of the solitary locusts reached a certain density, their eggs would hatch into young that were gregarious and would gather in large groups. These juveniles would then mature into the migratory adults that would form enormous swarms and leave the mountain valleys for the Great Plains of Canada and the United States. Eggs were deposited by the females as the swarms swept over the plains. The young that hatched from these eggs were also migratory and after acquiring wings, gathered into swarms that returned to the mountain valleys from which their parents had come. Here in their ancestral home they would lay eggs that would again produce the solitary form of their species. Although numerous specimens were collected during the plagues of the nineteenth century, only nine specimens are preserved from Texas. Hundreds of specimens of the Rocky Mountain locust have recently been taken from melting glaciers in the western United States. A comparison of the DNA of these glacial specimens with that of living species indicates that their closest relative is Bruner's spurthroated

[53] Jeffrey A. Lockwood, *Locust: The Devastating Rise and Mysterious Disappearance of the Insect That Shaped the American Frontier* (New York: Basic Books, 2004), 128.

grasshopper (*Melanoplus bruneri*), a resident of alpine meadows in the United States and Canada.[54]

The mass migrations of the Rocky Mountain locust were probably brought to a halt by the destruction of habitat in their permanent home. Pioneers moving into the Rocky Mountains during the 1870s and 1880s settled in the same areas occupied by the solitary form of the locust. Plowing destroyed the native grasses and the eggs of the locusts. Livestock competed with the locusts for available forage. Irrigation of land along streams killed the eggs of locusts. It is estimated that the breeding area within the valleys occupied by the locusts consisted of only a few thousand acres, an area that could have been quickly transformed by the activities of man and his domestic animals. Reduction in numbers of the solitary form resulted in the cessation of mass migrations. Ultimately, with their habitat destroyed, the species became extinct. Although most entomologists believe that the Rocky Mountain locust is truly extinct, there is speculation that isolated populations may yet exist in the Snake River Basin of Idaho and in Yellowstone National Park.[55] Thus far, specimens have not been produced and definitive proof is lacking.

The impact of the Rocky Mountain locust on Texans during the thirty-five years (1845–1880) that it periodically visited the Lone Star State is difficult to evaluate. Observers of the immense swarms experienced a sense of awe and helplessness as they desperately tried to cope with this mighty force of nature. Stockmen, farmers, nurserymen, and gardeners undoubtedly suffered economically from the destruction of livestock forage, fall gardens, orchards, and early spring crops. However, although the damage caused by the locusts was often severe, it was generally localized and limited to the central prairies while other regions of the state were relatively unaffected. The now extinct Rocky Mountain locust is perhaps best remembered as an agricultural pest that presented an ephemeral yet significant challenge to the endurance and ingenuity of rural Texans struggling to make the land bring forth its fruit.

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[54] Ibid., 137–158; A. B. Gurney and A. R. Brooks, "Grasshoppers of the Mexicanus Group, Genus *Melanoplus* (Orthoptera: Acrididae)," *Proceedings U.S. National Museum*, 110 (1959), 1–93. There are reportedly only 487 specimens of the Rocky Mountain locust found in all North American collections; Lockwood, *Locust*, 183. Jeffrey A. Lockwood, R. A. Nunamaker, R. E. Pfadt, and L. D. DeBrey, "Grasshopper Glacier: Characterization of a Vanishing Biological Resource," *American Entomologist*, 36 (1990), 18–27. Lockwood, *Locust*, 209–224.

[55] Jeffrey A. Lockwood and L. D. Debrey, "A Solution for the Sudden and Unexplained Extinction of the Rocky Mountain Grasshopper," *Environmental Entomology*, 19 (1990), 1194–1205; Lockwood, *Locust*, 225–254. The idea that the Rocky Mountain locust would eventually be controlled by modification of its habitat in the Rocky Mountains was originally suggested by Riley, et al., *First Annual Report of the United States Entomological Commission*, 126–127. C. K. Yoon, "Looking Back at the Days of the Locust," *New York Times*, Apr. 23, 2002; Lockwood, *Locust*, 262.